How an Email Travels the Internet

The Journey of an Email

- Emails are addressed and sent to one another based on a naming and routing system the Internet understands.
- Just as the United States Postal Service has a system for identifying people at particular locations, so does the Internet for email.

The Journey of an Email

- Components of a U.S. mailing address
  - One identifies a particular person.
  - The other identifies the location of that person in the United States.

John Doe
123 Somewhere Rd
Sacramento, CA 95831

The Journey of an Email

- The components of an email address are similar in that they identify:
  - A person, and
  - The location of that person on the Internet.

john.doe@somewhere.com

- The @ symbol is used to separate the components.

The Journey of an Email

- The information on the left of the @ is called the user ID.
  - America Online (AOL) calls the user ID the user's Screen Name.
  - In general, no two users have the same user ID at the same location.
    - Home users will sometimes share a user ID.
      - E.g., one for the entire family.

The Journey of an Email

- A person obtains a user ID when it is either assigned to them or they voluntarily create one.
- Assigned IDs
  - Places of employment
  - Often the employee's name or combination of first and last name, for example: john.smith@search.org

The Journey of an Email

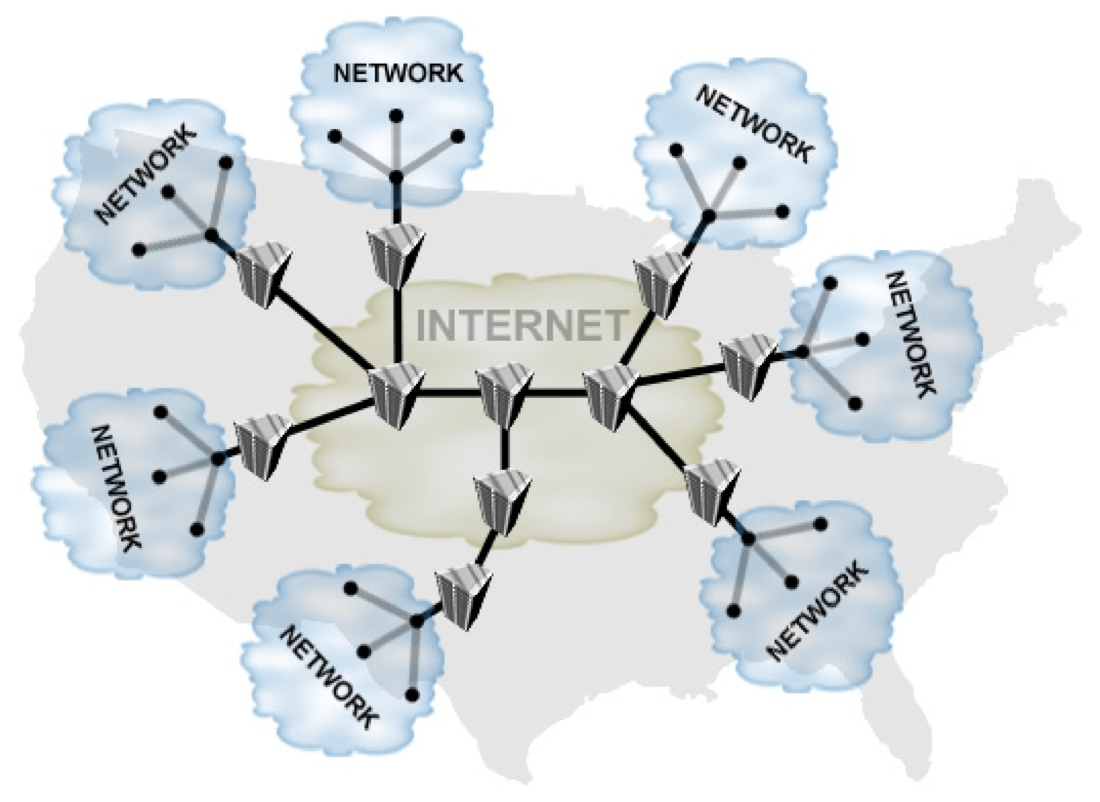
- User-chosen IDs (voluntarily created)
  - Internet Service Providers (ISPs)
    - Customer chooses own unique user ID.
    - Can have multiple unique user IDs.
  - Free, online, email service providers.
    - Examples: Hotmail and Yahoo!
  - Users often choose IDs that describe them, their interests, location, or vocation. Examples:
    - thejohnsons@localhost.com
    - soccerfan@acmeisp.com
    - boston17@hotmail.com
    - petshopboys@aol.com

The Journey of an Email

- The information to the right of the @ that identifies the location of the person on the Internet is called the domain name or domain information.
  - Not a physical location on the Internet.
  - Identifies a network.

The Journey of an Email

- The Internet is really a giant network of computer networks that span not only the United States, but also the entire globe.



The Journey of an Email

- Networks are identified on the Internet by both names and numbers.
  - Domain names makes it easier for people to remember which network is for what.
    - petshopboys@aol.com identifies the user "petshopboys" at the aol.com network.
  - The Internet uses numbers assigned to networks to get information from one network to another – not the domain name.

The Journey of an Email

- The Internet numbers also identify individual computers.
  - Every computer connected to the Internet must have one of these unique numbers.
- The numbers are called IP (Internet Protocol) addresses and are a series of four numbers separated by periods. For example:
  - 64.162.18.35
  - 192.168.1.74

The Journey of an Email

- IP addresses serve the same purpose as phone numbers for the telephone system.
  - They uniquely identify every device connected to the network.
- But when people send emails to one another, they do not use IP addresses. They use the recipient's user ID and network information – the email address.

The Journey of an Email

- This is possible because of the Domain Name System (DNS) on the Internet that serves a purpose similar to that of phonebooks.
  - Phonebooks show the link between people’s names and the meaningless phone number assigned to them.
  - DNS is a record of the link between domain names and the meaningless IP addresses assigned to them.

The Journey of an Email

- The Domain Name System operates on computers called Domain Name Servers and Nameservers.
- When an email is sent, the Domain Name System is used to translate the network name (petshopboys@aol.com) to an IP address.
  - The IP address is the address of the email server for that network.
  - Email servers are like electronic post offices – they send and receive electronic mail.

The Journey of an Email

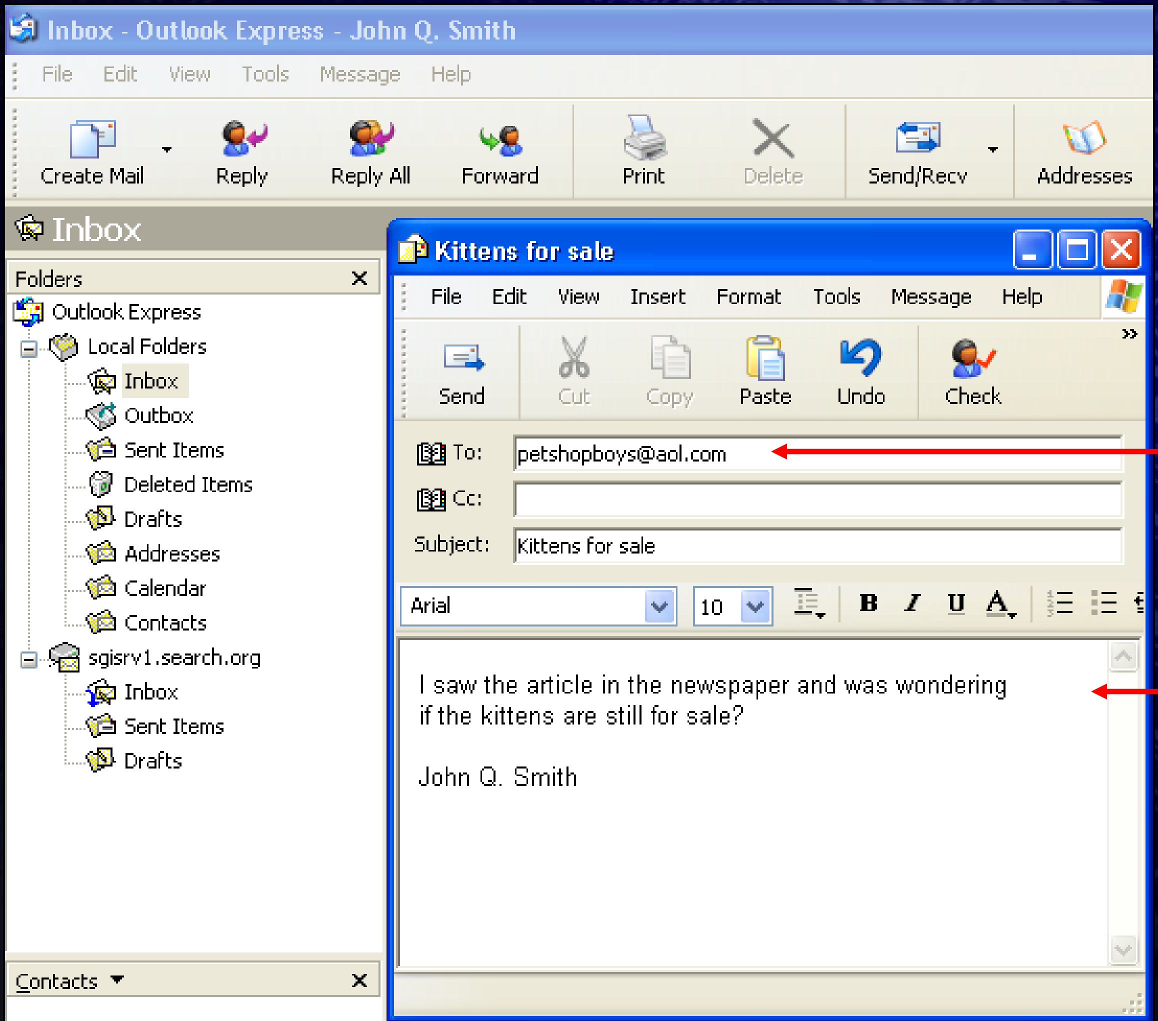
- Example of the process of sending an email.
  1. The sender enters the message and receiver's email address in their email program, then sends it.

The Journey of an Email

- Two examples of popular email programs are:
  - Outlook Express
  - America Online (AOL)

The Journey of an Email

- Outlook Express:

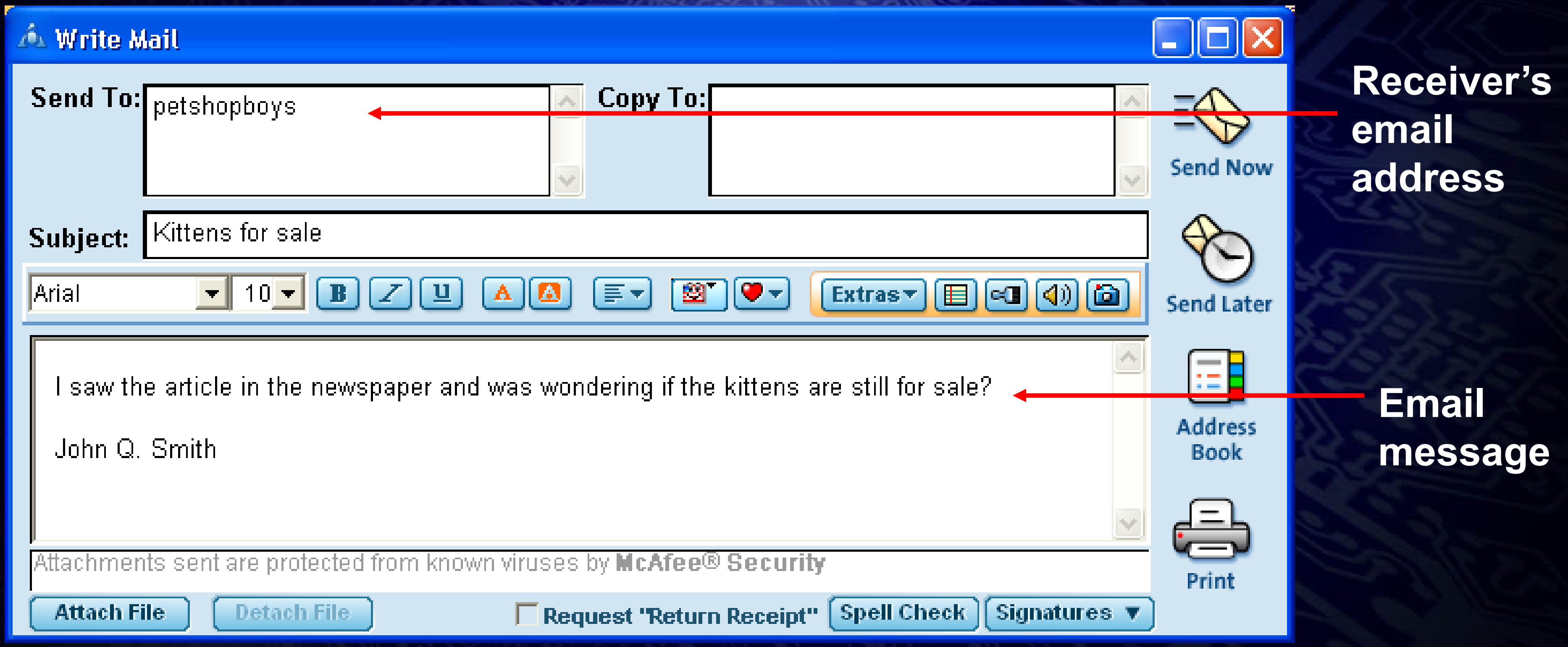


Receiver's email address

Email message

The Journey of an Email

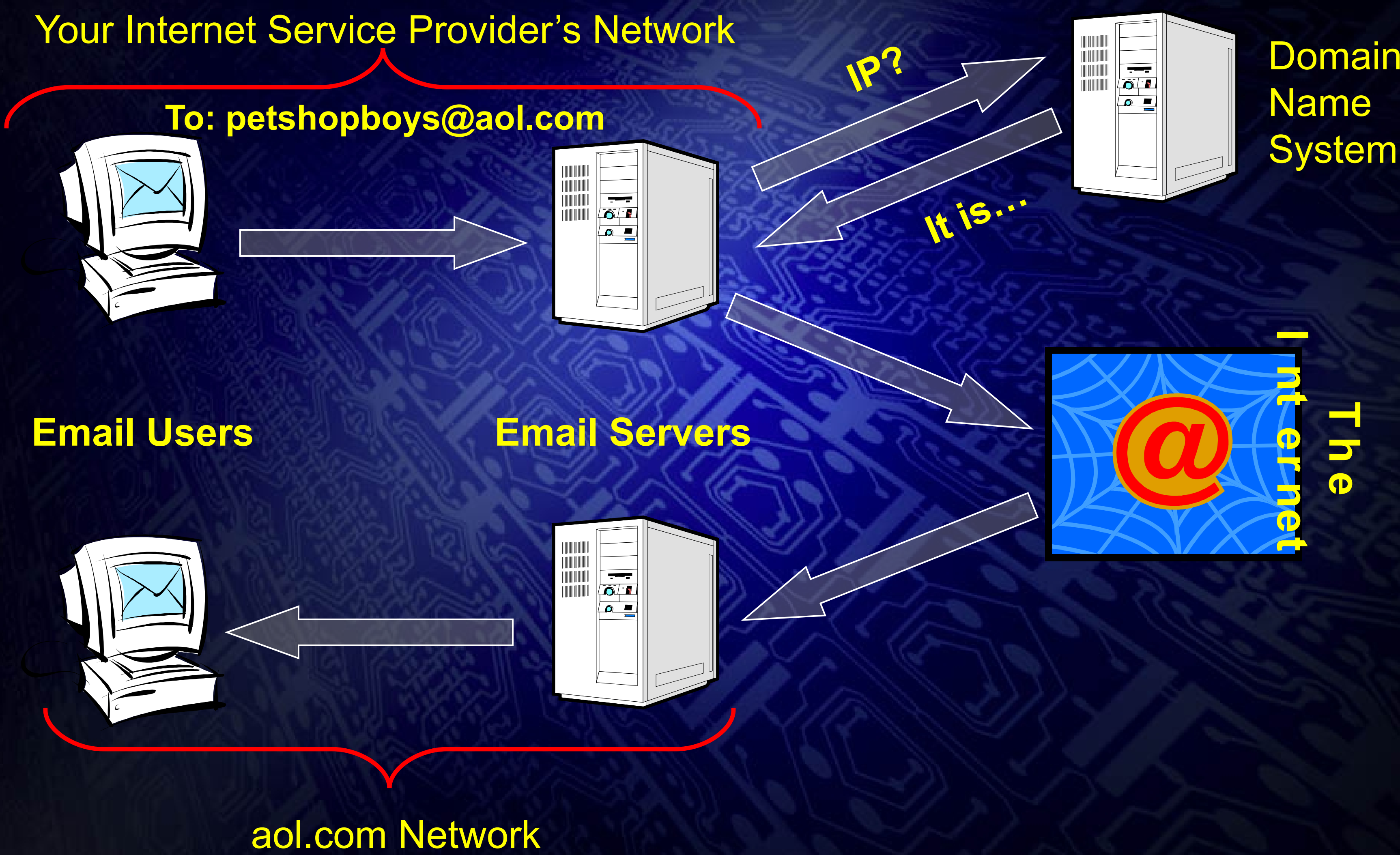
- America Online (AOL):



The Journey of an Email

- Example of the process of sending an email (continued).
  2. The email server for the sender's network uses the Domain Name System to determine the IP address of the email server of the receiver's network.
  3. The email is sent across the Internet to the IP address of the receiver's email server.

The Journey of an Email



Example of an email's journey

The Journey of an Email

- An abbreviated record of an email's journey through the Internet is recorded in a document called an email header.
  - The email header is transmitted with the email message.
- Investigators can use the email header to determine who it is that sent a message.
  - This process is called email tracing.

Email Records

- Example of an email header:

Return-Path: jqs@search.org
Received: from rly-xa05.mx.aol.com (rly-xa05.mail.aol.com [172.20.105.74])
by air-xa04.mail.aol.com (v89.12) with ESMTP id MAILINXA42-1014220440;
Mon, 14 Oct 2002 22:04:40 –0400
Received: from mail.search.org ([64.168.6.142])
by rly-xa05.mx.aol.com (v89.12) with ESMTP id MAILRELAYINXA57-1014220425;
Mon, 14 Oct 2002 22:04:25 –0400
Received: from mail ([64.168.6.141]) by mailserver.search.org with Microsoft SMTPSVC(5.0.2195.4905);
Mon, 14 Oct 2002 19:04:05 –0700
Message-ID: 000a01c273ef$23ff7f40$8206a840@mail.search.org
From: "John Q. Smith" jqs@search.org
To: petshopboys@aol.com
Subject: Kittens for sale
Date: Mon, 14 Oct 2002 19:02:24 –0700
MIME-Version: 1.0
Content-Type: multipart/alternative;
boundary="----=_NextPart_000_0005_01C273B4.3B06BCC0“
X-Priority: 3
X-MSMail-Priority: Normal
X-Mailer: Microsoft Outlook Express 6.00.2800.1106
X-MimeOLE: Produced By Microsoft MimeOLE V6.00.2800.1106
X-OriginalArrivalTime: 15 Oct 2002 02:04:05.0828 (UTC) FILETIME=[2401F040:01C273EF]

Email Records

- Important information for tracing.

Return-Path: jqs@search.org
Received: from rly-xa05.mx.aol.com (rly-xa05.mail.aol.com [172.20.105.74])
by air-xa04.mail.aol.com (v89.12) with ESMTP id MAILINXA42-1014220440;
Mon, 14 Oct 2002 22:04:40 –0400
Received: from mail.search.org ([64.168.6.142])
by rly-xa05.mx.aol.com (v89.12) with ESMTP id MAILRELAYINXA57-1014220425;
Mon, 14 Oct 2002 22:04:25 –0400
Received: from **mail ([64.168.6.141])** by mailserver.search.org with Microsoft SMTPSVC(5.0.2195.4905);
Mon, 14 Oct 2002 19:04:05 –0700
Message-ID: 000a01c273ef$23ff7f40$8206a840@mail.search.org
From: "John Q. Smith" jqs@search.org
To: petshopboys@aol.com
Subject: Kittens for sale
Date: Mon, 14 Oct 2002 19:02:24 –0700
MIME-Version: 1.0
Content-Type: multipart/alternative;
boundary="----=_NextPart_000_0005_01C273B4.3B06BCC0“
X-Priority: 3
X-MSMail-Priority: Normal
X-Mailer: Microsoft Outlook Express 6.00.2800.1106
X-MimeOLE: Produced By Microsoft MimeOLE V6.00.2800.1106
X-OriginalArrivalTime: 15 Oct 2002 02:04:05.0828 (UTC) FILETIME=[2401F040:01C273EF]

Will sometimes identify the sender’s email server

Email Records

- Example of an email server log file (from the sender's email server).
  - c-ip = sender's IP address.
  - cs-username = hostname of sender's computer.

C	D	E	F	G
c-ip	cs-username	s-computername	s-ip	cs-uri-query
64.168.6.152	jqsmith	MAIL	64.168.6.141	+<000a01c273ef$23ff7f40$820
64.168.6.152	jqsmith	MAIL	64.168.6.141	mail